

Astro-1_down_AA BP

nuclear-transcribed mRNA poly(A) tail shortening	piRNA processing	nuclear-transcribed mRNA catabolic process, nonsense-mediated decay	mRNA destabilization	blastocyst formation	dorsal/ventral neural tube patterning	in utero embryonic development
nuclear-transcribed mRNA catabolic process, deadenylation-dependent decay	RNA 3'-end processing	RNA destabilization	nuclear-transcribed mRNA catabolic process	dorsal/ventral neural tube patterning morphogenesis of embryonic epithelium		
positive regulation of mRNA catabolic process	positive regulation of mRNA metabolic process	regulation of mRNA stability	regulation of mRNA catabolic process	blastocyst development	neural tube development	
regulatory ncRNA processing	regulatory ncRNA-mediated gene silencing	regulation of RNA stability	mRNA catabolic process	neural tube formation	embryonic epithelial tube formation	protein localization to ciliary transition zone
cell differentiation in spinal cord	spinal cord motor neuron differentiation	central nervous system neuron development	spinal cord development	neural tube formation	embryonic epithelial tube formation	protein localization to ciliary transition zone
central nervous system neuron axonogenesis	ventral spinal cord development	central nervous system neuron differentiation	telencephalon development	regulation of smoothed signaling pathway	dorsal/ventral pattern formation	

Astro-1_up_AA CC

postsynaptic density membrane		postsynaptic specialization membrane		asymmetric synapse		postsynaptic density		perikaryon					
synaptic membrane		postsynaptic specialization		neuron to neuron synapse		presynaptic endocytic zone membrane		dendritic perikaryon					
						photoreceptor ribbon synapse		neuronal cell body					
postsynaptic membrane		presynaptic membrane		Schaffer collateral – CA1 synapse		presynaptic endocytic zone		neuronal cell body membrane					
				GABA-ergic synapse		ribbon synapse		endocytic vesicle					
cation channel complex		AMPA glutamate receptor complex		ionotropic glutamate receptor complex		voltage-gated calcium channel complex		calcium channel complex		voltage-gated potassium channel complex		endocytic vesicle	
												endocytic vesicle membrane	
monoatomic ion channel complex		transmembrane transporter complex		transporter complex		neurotransmitter receptor complex		potassium channel complex		L-type voltage-gated calcium channel complex		axolemma	
								plasma membrane signaling receptor complex				synaptic cleft	

Astro-1_up_AA BP



Astro-1_up_AA MF

voltage-gated channel activity		voltage-gated monoatomic cation channel activity		voltage-gated monoatomic ion channel activity		metal ion transmembrane transporter activity		phosphatidylinositol bisphosphate binding	
								phosphatidylinositol biphosphate binding	
gated channel activity	monoatomic ion channel activity		voltage-gated potassium channel activity		potassium channel activity		proline:sodium symporter activity		phosphatidylinositol phosphate binding
monoatomic cation channel activity	voltage-gated channel activity		calcium channel activity		calcium ion transmembrane transporter activity		neutral L-amino acid:sodium symporter activity		phosphatidate phosphatase activity
	passive transmembrane transporter activity		potassium ion transmembrane transporter activity		voltage-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		amino acid:sodium symporter activity		phosphatidate phosphatase activity
delayed rectifier potassium channel activity	voltage-gated calcium channel activity		branched-chain amino acid transmembrane transporter activity		neurotransmitter transmembrane transporter activity		amino acid:monoatomic cation symporter activity		lipid phosphatase activity
AP-1 adaptor complex binding		AP-1 adaptor complex binding		AP-1 adaptor complex binding		AP-1 adaptor complex binding		channel regulator activity	

Astro-1_up_ALL CC

G protein-coupled receptor complex

postsynaptic membrane

GABA receptor complex

synaptic membrane

GABA receptor complex

synaptic membrane

Astro-1_up_ALL BP

negative regulation of
adenylate cyclase activity

negative regulation of lyase activity

negative regulation of lyase activity

regulation
of adenylate
cyclase
activity

synaptic vesicle docking

synaptic vesicle docking

vesicle docking
involved in exocytosis

neuron-glia cell signaling

neuron-glia cell signaling

gamma-aminobutyric
acid signaling pathway

lipid import into cell

Astro-1_up_ALL MF



Astro-2_up_ALL MF



Astro-3_down_ALL CC

myelin sheath

myelin sheath

compact myelin

Astro-3_down_ALL BP

axon ensheathment

ensheathment of neurons

iron ion
transport

myelination

**iron ion
transport**

myelination

myelin assembly

regulation of
vesicle-mediated
transport

substantia nigra development

Astro-3_down_ALL MF



Astro-3_up_ALL CC



Astro-3_up_ALL BP



Astro-3_up_ALL MF



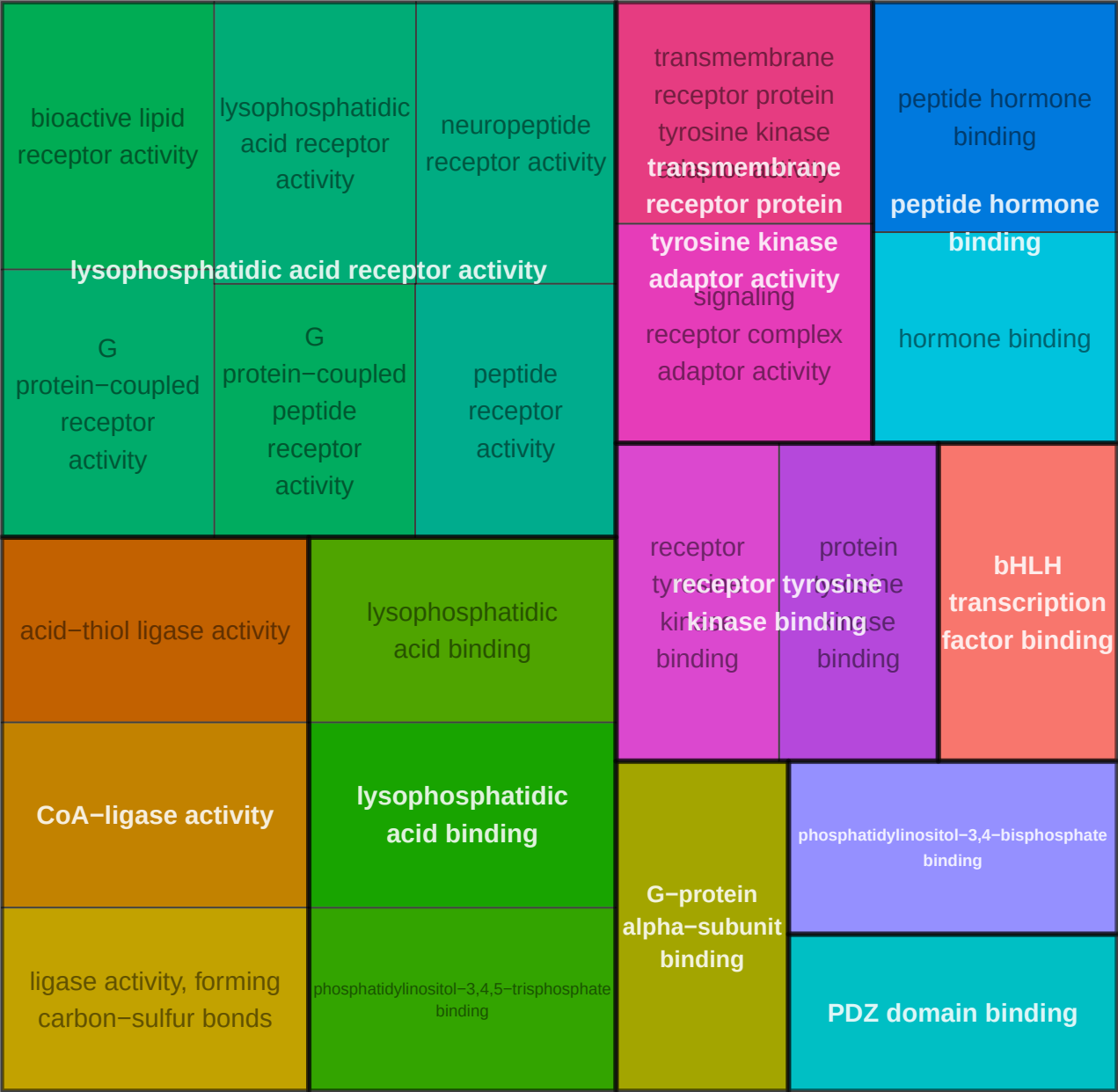
Excit-L2_5-1_down_ALL CC

dendritic spine

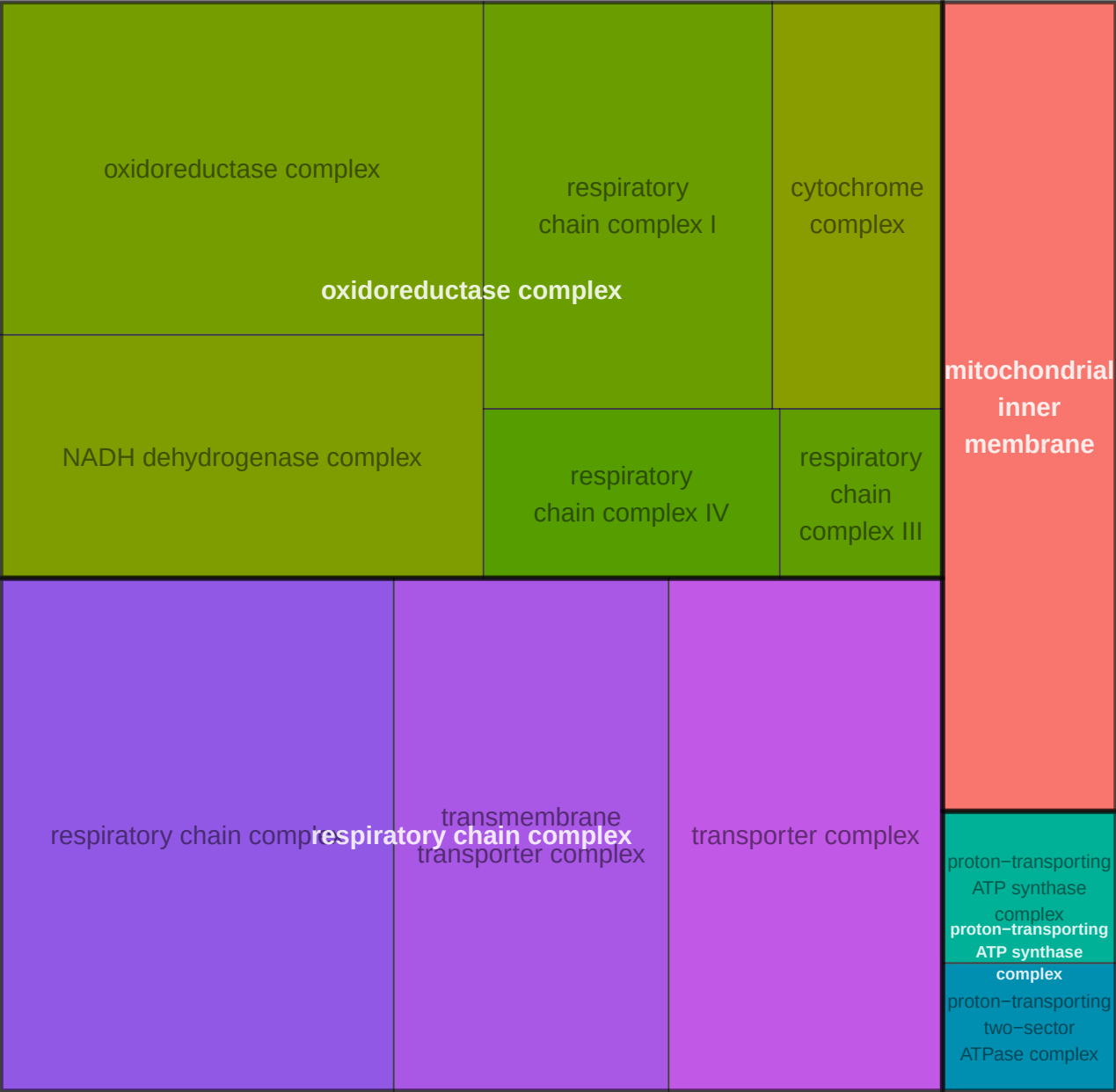
neuron spine

neuron spine

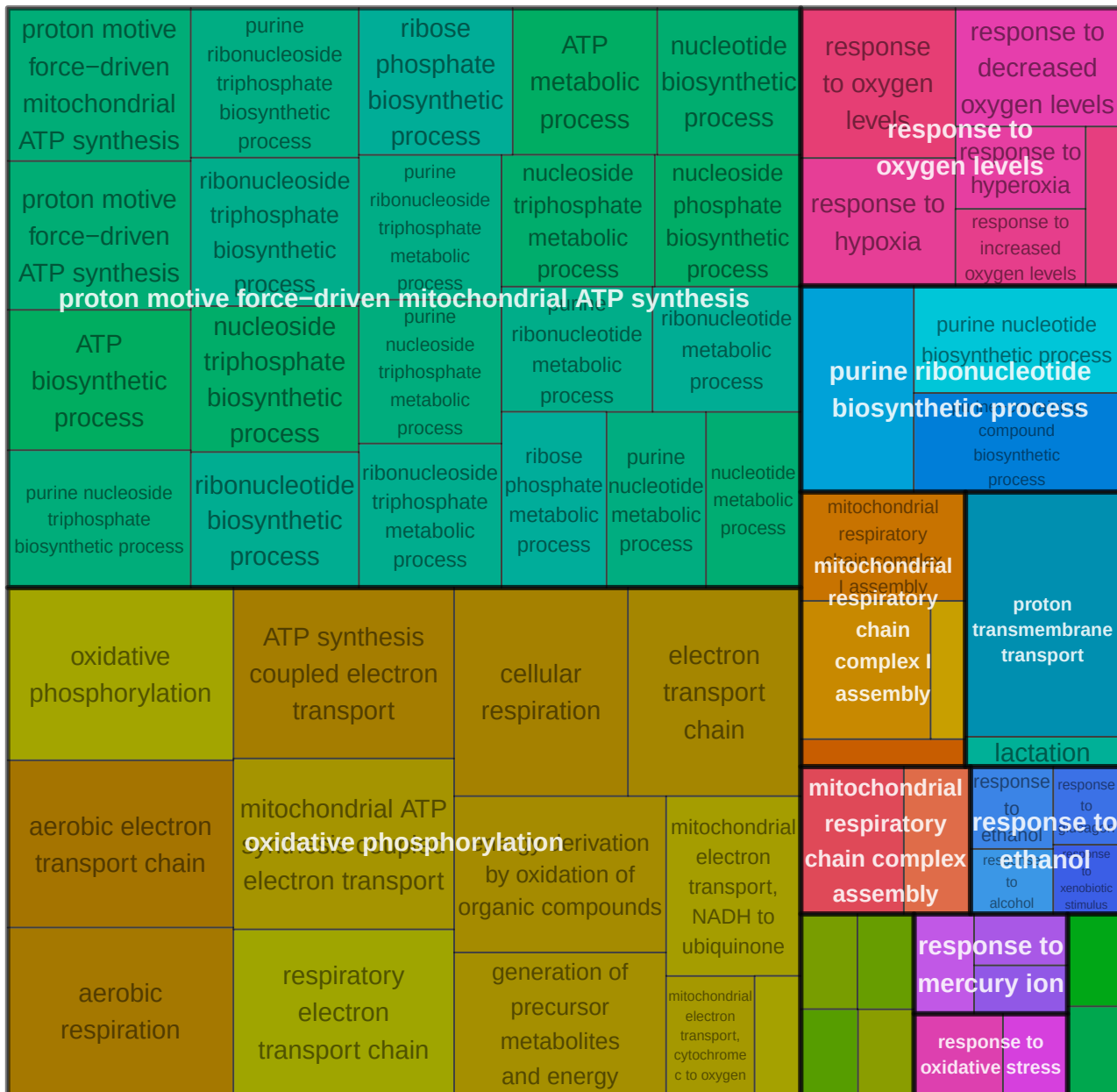
Excit-L2_5-1_down_ALL MF



Excit-L5-2_down_ALL CC



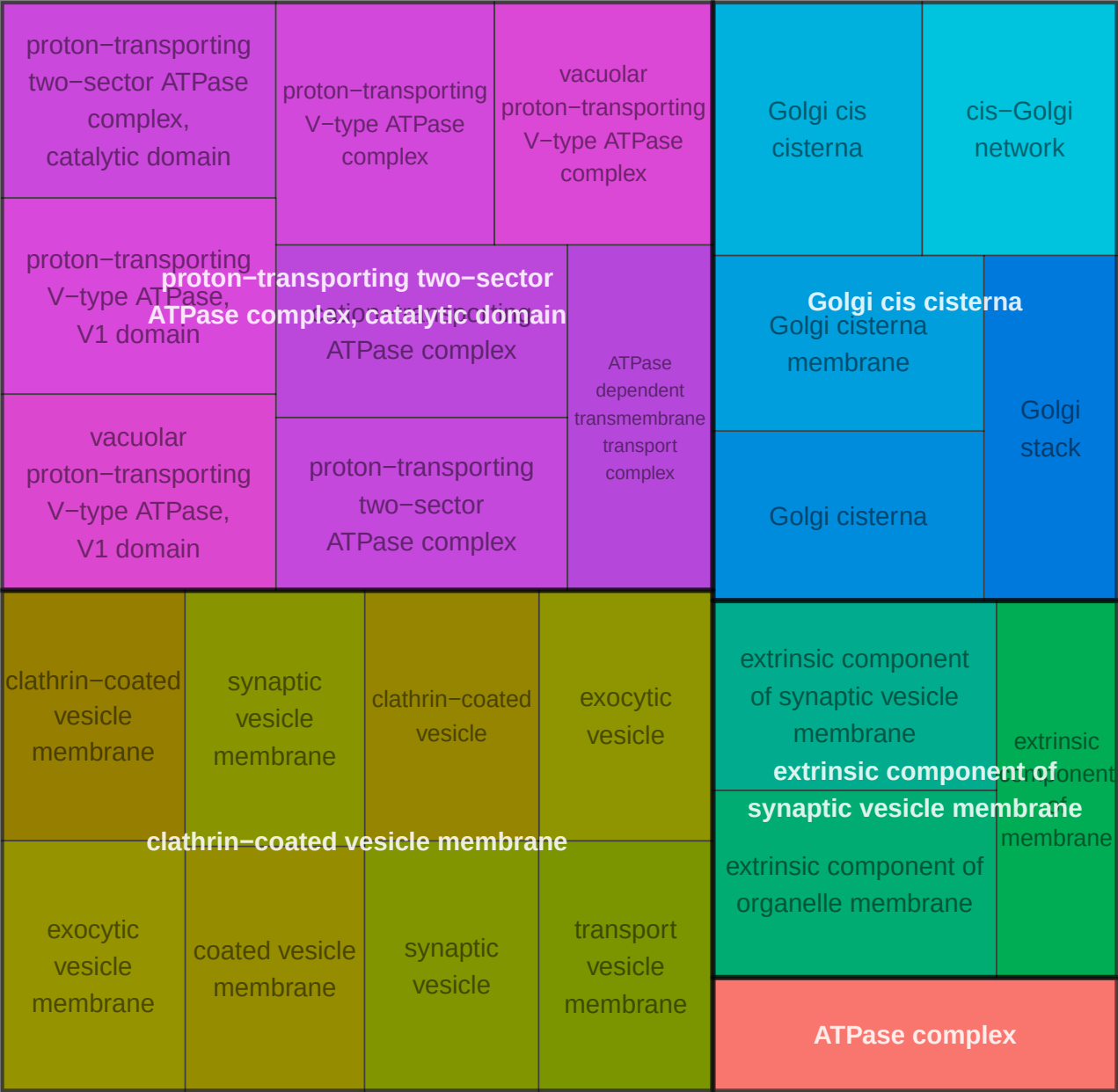
Excit-L5-2 down ALL BP



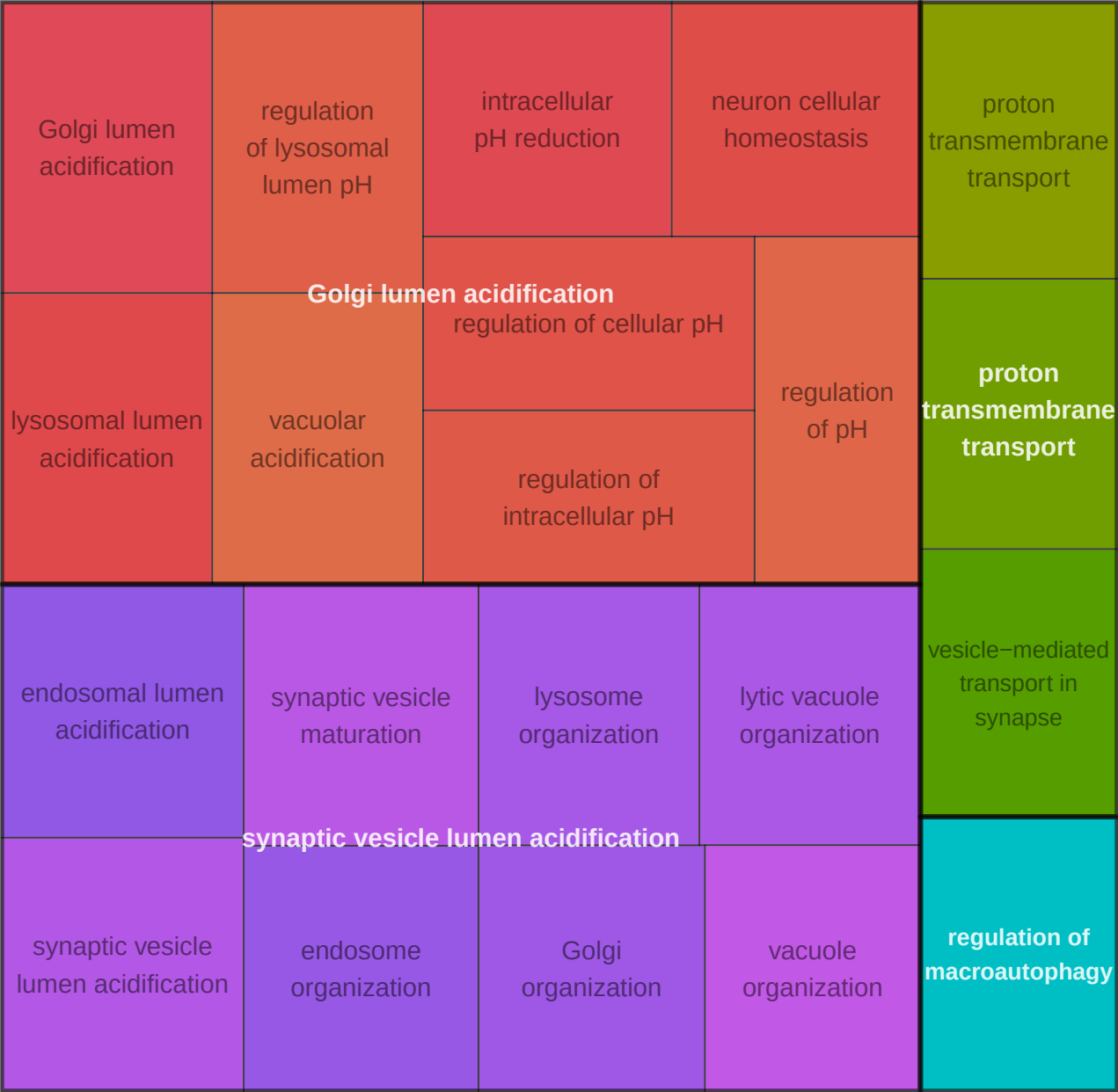
Excit-L5-2_down_ALL MF



Excit-L5-2_up_ALL CC



Excit-L5-2_up_ALL BP



Inhib-Sst_down_ALL CC

cytosolic large ribosomal subunit

large ribosomal subunit

cytosolic ribosome

ribosomal subunit

cytosolic ribosome

ribosome

Inhib-Vip_down_ALL BP

axon ensheathment in
central nervous system

axon ensheathment in central nervous system

Macro_down_ALL CC



Macro_down_ALL BP



Macro_down_ALL MF



Macro up AA BP



Macro_up_AA MF

dopamine receptor binding

beta-catenin binding

chromatin DNA binding

G-protein
beta/gamma-subunit
complex binding

transcription coregulator binding

Macro_up_ALL MF

carbohydrate
phosphatase activity

**fructose-2,6-bisphosphate
2-phosphatase activity**

fructose-2,6-bisphosphate
2-phosphatase activity

tumor necrosis factor
receptor activity

tumor necrosis factor receptor activity

sugar-phosphatase activity

**phosphofructokinase
activity**

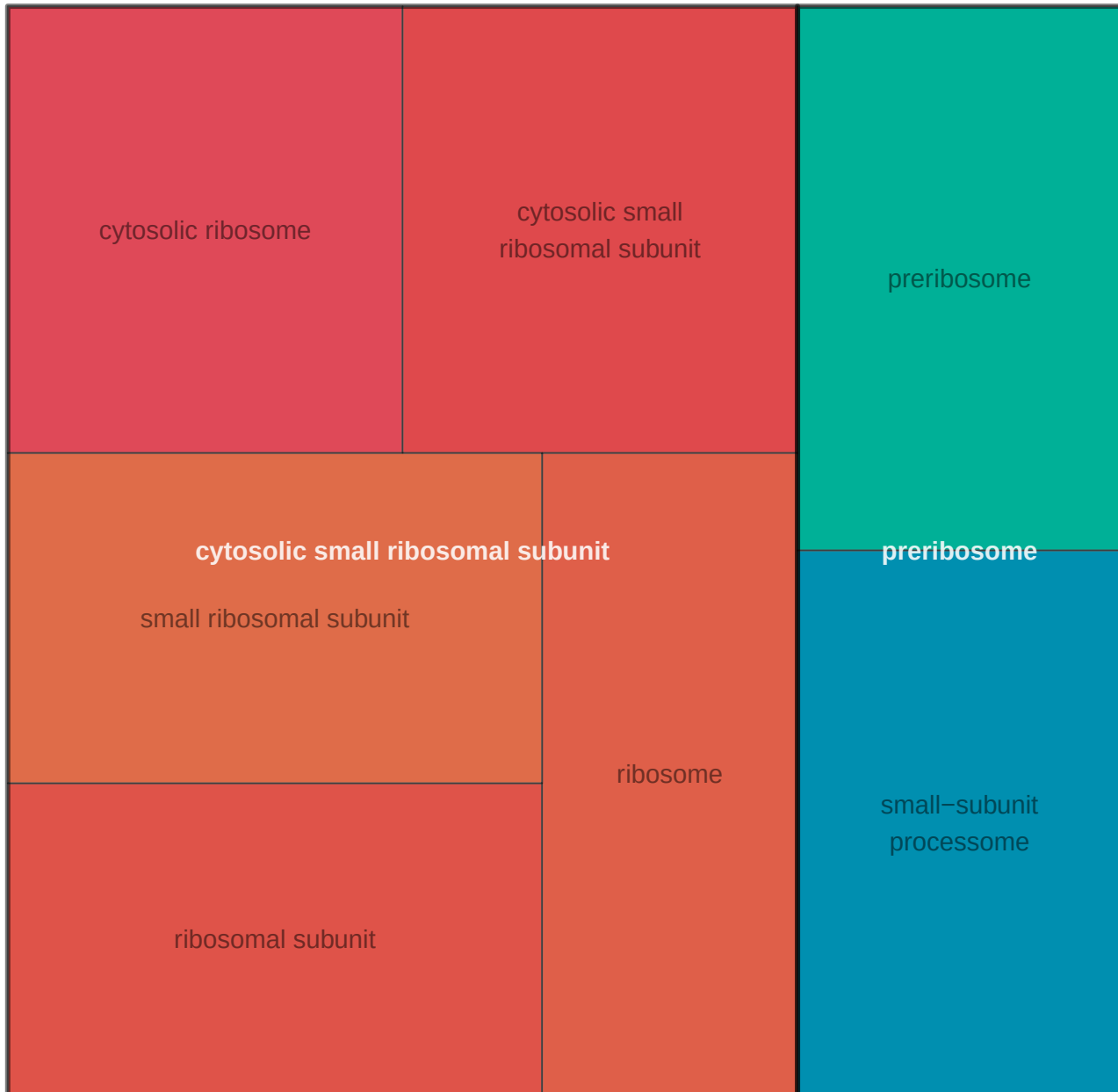
**transcription regulator
activator activity**

tumor necrosis factor binding

tumor necrosis
factor binding

dopamine receptor binding

Micro-1_down_AA CC



Micro-1_down_AA BP

negative regulation of post-translational protein modification	negative regulation of protein modification by small protein conjugation or removal	negative regulation of protein ubiquitination	neural tube closure	primary neural tube formation	positive regulation of intrinsic apoptotic signaling pathway by p53 class mediator	positive regulation of signal transduction by p53 class mediator
negative regulation of proteolysis involved in protein catabolic process	negative regulation of ubiquitin-dependent protein catabolic process	regulation of ubiquitin-dependent protein catabolic process	tube closure	embryonic epithelial tube formation	positive regulation of intrinsic apoptotic signaling pathway by p53 class mediator	
negative regulation of ubiquitin-dependent protein catabolic process		regulation of protein modification by small protein conjugation or removal	epithelial tube formation	tube formation	regulation of signal transduction by p53 class mediator	signal transduction by p53 class mediator
negative regulation of proteolysis		regulation of proteolysis involved in protein catabolic process	neural tube formation			
negative regulation of proteolysis	regulation of post-translational protein modification	negative regulation of protein modification process	morphogenesis of embryonic epithelium	mesenchymal cell differentiation	neural crest cell differentiation	
negative regulation of molecular function	regulation of ubiquitin protein ligase activity	regulation of ubiquitin-protein transferase activity	morphogenesis of embryonic epithelium			
negative regulation of protein ligase activity	negative regulation of NF-kappaB transcription factor activity	negative regulation of transferase activity	neural tube development	mesenchyme development		rRNA processing
negative regulation of ubiquitin-protein transferase activity	negative regulation of DNA-binding transcription factor activity	negative regulation of catalytic activity	positive regulation of intrinsic apoptotic signaling pathway	regulation of intrinsic apoptotic signaling pathway	rRNA processing	rRNA processing
			positive regulation of apoptotic signaling pathway	apoptotic signaling pathway	ribosomal small subunit biogenesis	protein stabilization

Micro-1_down_AA MF

ubiquitin ligase inhibitor activity

ubiquitin-protein transferase regulator activity

ubiquitin-protein
transferase
inhibitor activity

ubiquitin-protein
transferase
regulator activity

NF-kappaB binding

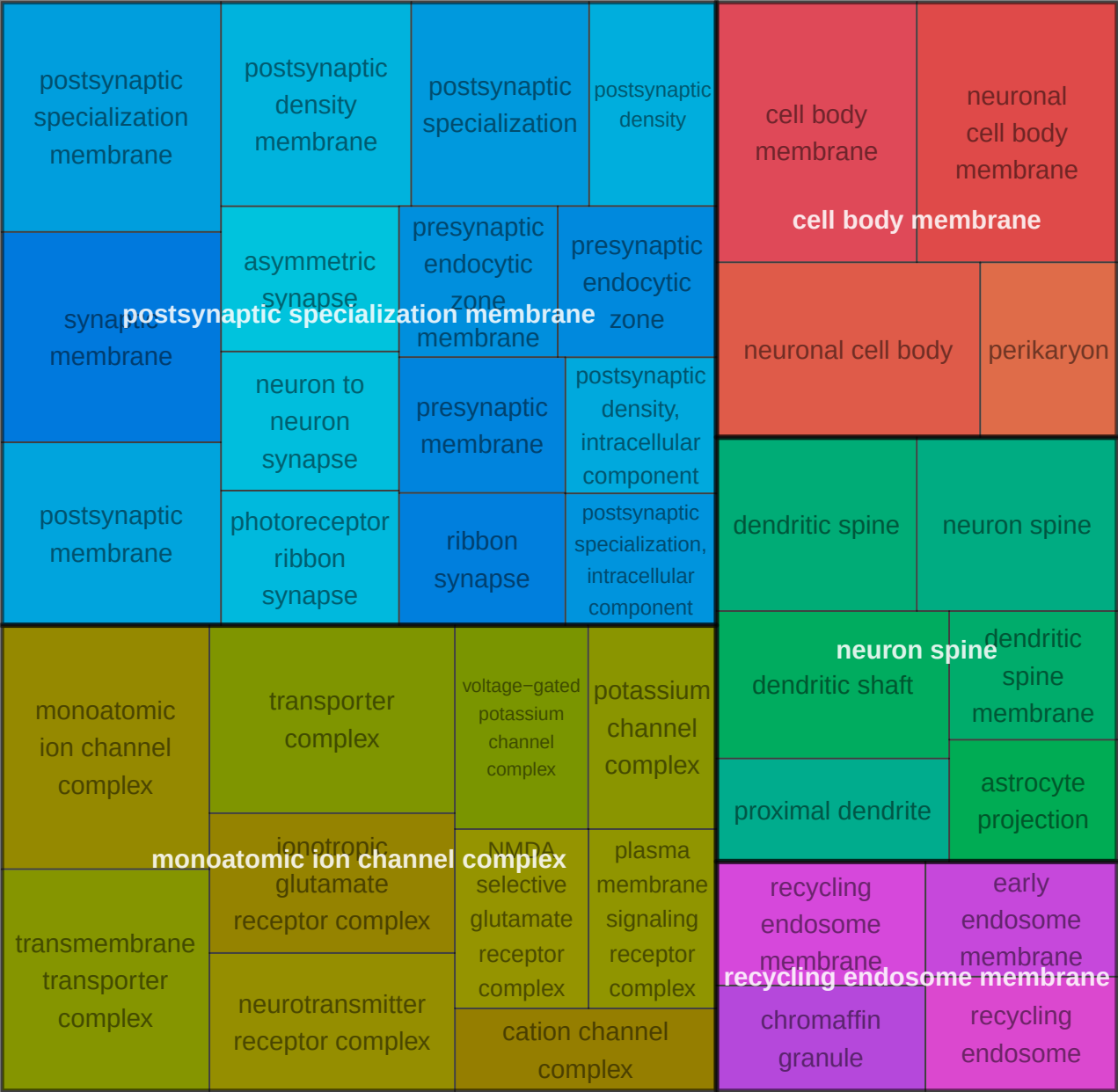
mRNA 5'-UTR binding

mRNA 5'-UTR binding

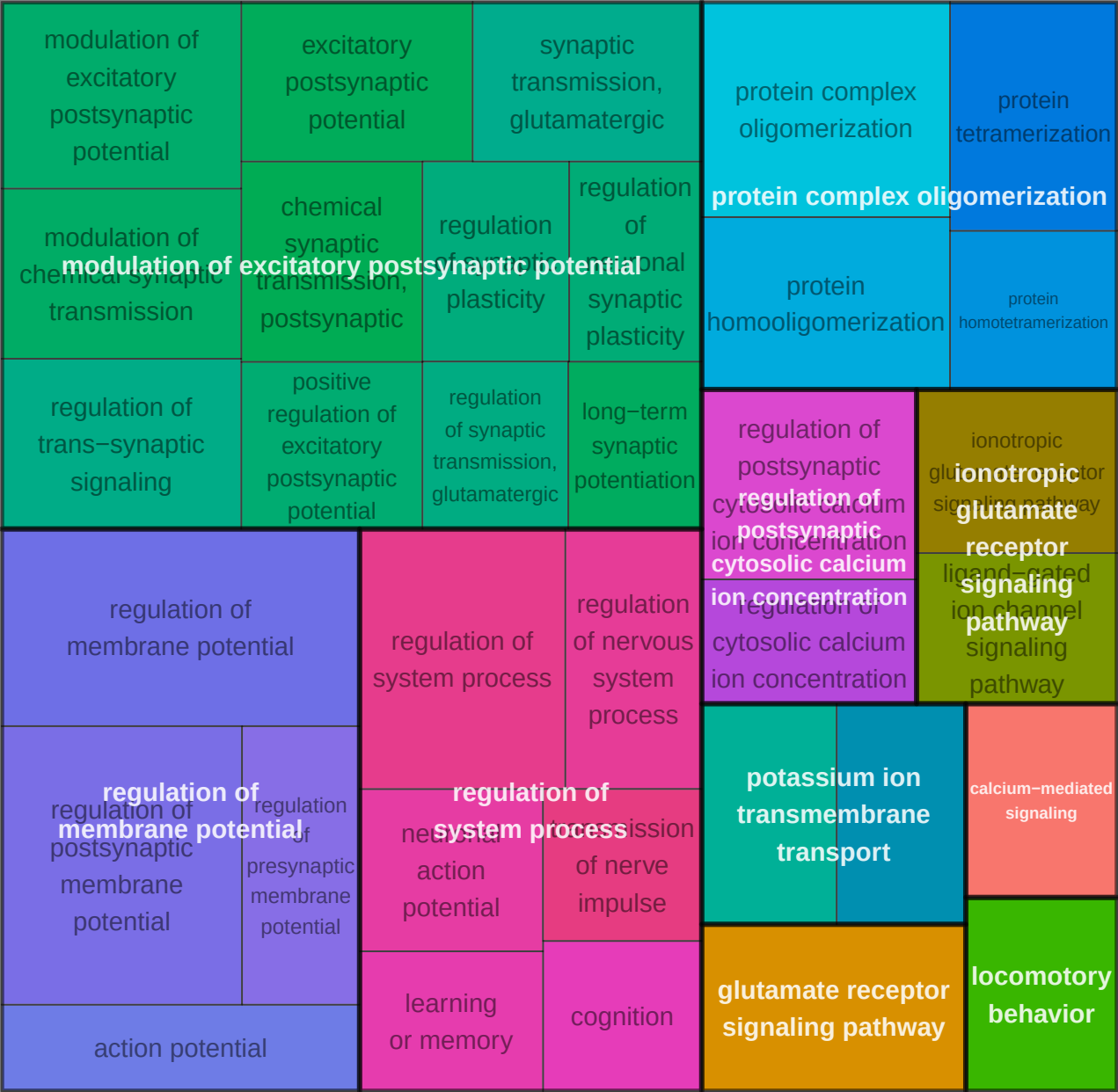
mRNA 3'-UTR binding

**RNA
methyltransferase
activity**

Micro-1_up_AA CC



Micro-1_up_AA BP



Micro-1_up_AA MF

gated channel activity	metal ion transmembrane transporter activity	monoatomic ion channel activity	channel activity		passive transmembrane transporter activity		adenosine receptor binding		beta-2 adrenergic receptor binding			
	glutamate-gated calcium ion channel activity		ligand-gated monoatomic cation channel activity		voltage-gated monoatomic cation channel activity		potassium ion transmembrane transporter activity		adenosine receptor binding			
monoatomic cation channel activity	voltage-gated potassium channel activity	gated channel activity	ligand-gated calcium channel activity		extracellular ligand-gated monoatomic ion channel activity	amino acid transmembrane transporter activity	calcium channel activity		peptide hormone receptor binding	hormone receptor binding		
voltage-gated channel activity			ligand-gated channel activity		calcium ion transmembrane transporter activity	organic acid transmembrane transporter activity	organic anion transmembrane transporter activity	carboxylic acid binding		organic acid binding		
voltage-gated monoatomic ion channel activity	ligand-gated channel activity		glutamate binding					glycine binding				
	ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		amide binding		amide binding			amyloid-beta binding		amyloid-beta binding		
voltage-gated monoatomic ion channel activity	potassium channel activity		ligand-gated monoatomic ion channel activity		carboxylic acid transmembrane transporter activity	delayed rectifier potassium channel activity		outward rectifier potassium channel activity		amide binding		peptide binding
		potassium channel activity		CoA carboxylase activity		adenylate cyclase activity		adenylate cyclase activity				
glutamate receptor activity	neurotransmitter receptor activity	glutamate receptor activity	transmitter-gated channel activity	AMPA glutamate receptor activity		G protein-coupled receptor activity involved in regulation of postsynaptic membrane potential		CoA carboxylase activity		adenylate cyclase binding		
postsynaptic neurotransmitter receptor activity	glutamate-gated receptor activity			NMDA glutamate receptor activity				immunoglobulin binding		mRNA regulatory element binding repressor activity		
				adenylate cyclase inhibiting G protein-coupled glutamate receptor activity				myosin V binding		G-protein beta-subunit binding		

Micro-3_up_EA CC



Micro-3_up_EA BP



Micro-3_up_EA MF

GABA-gated chloride ion channel activity		extracellular ligand-gated monoatomic ion channel activity		chloride transmembrane transporter activity		monoatomic anion transmembrane transporter activity		adenylate cyclase regulator activity	
ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		ligand-gated monoatomic ion channel activity involved in regulation of presynaptic membrane potential		inorganic anion transmembrane transporter activity		ligand-gated monoatomic ion channel activity		cyclase regulator activity	
ligand-gated monoatomic anion channel activity				ligand-gated channel activity				cyclase regulator activity	
G protein-coupled neurotransmitter receptor activity		GABA-A receptor activity		neurotransmitter receptor activity involved in regulation of postsynaptic membrane potential		transmitter-gated channel activity		GABA receptor binding	
G protein-coupled receptor activity		GABA receptor activity		transmitter-gated monoatomic ion channel activity involved in regulation of postsynaptic membrane potential		postsynaptic neurotransmitter receptor activity		G-protein alpha-subunit binding	
				transmitter-gated monoatomic ion channel activity		neurotransmitter receptor activity			

Micro-4_up_ALL CC



Micro-4_up_ALL BP

modulation of chemical
synaptic transmission

regulation of trans-synaptic signaling

regulation of trans-synaptic signaling

Oligo-1_up_AA CC



Oligo-1_up_AA MF

P-type calcium
transporter activity

P-type ion
transporter
activity

calmodulin binding

P-type calcium transporter activity

P-type transmembrane
transporter activity

ATPase-coupled
monoatomic cation
transmembrane
transporter activity

glutamate receptor binding

calcium/calmodulin-dependent
protein kinase activity

syntaxin-1 binding

syntaxin-1 binding

diacylglycerol binding

Oligo-3_down_AA, Oligo-3_down_EA BP

mannosylation

protein mannosylation

**glycoprotein
biosynthetic
process**

protein O-linked mannosylation

mannosylation

glycosylation

macromolecule
glycosylation

protein O-linked glycosylation

protein glycosylation

**rhythmic
process**

calcium ion homeostasis

Oligo-3_down_ALL CC



Oligo-3_down_ALL BP

axon ensheathment

myelination

myelination

oligodendrocyte development

ensheathment of neurons

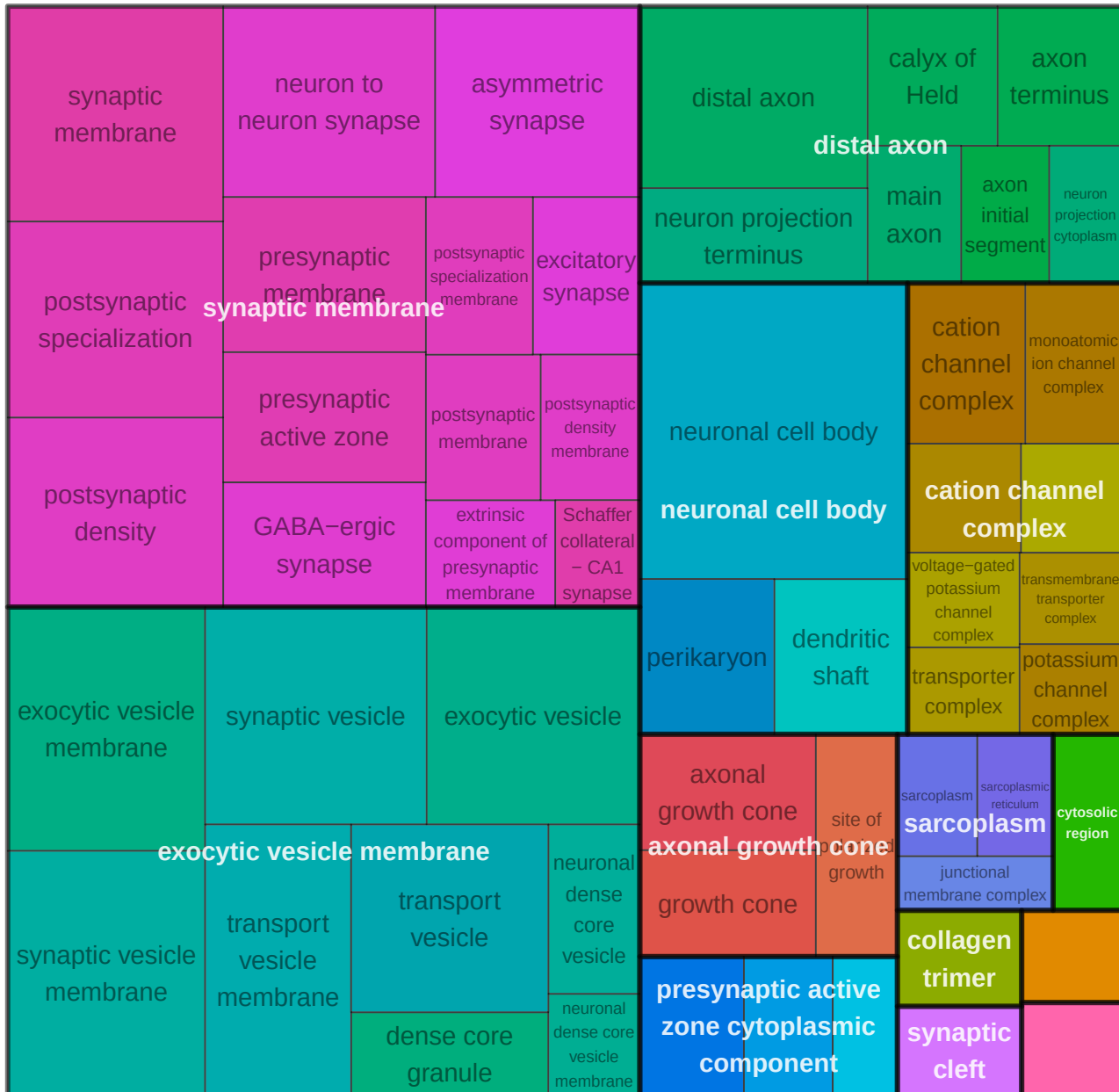
oligodendrocyte differentiation

Oligo-3_down_ALL MF

structural constituent of myelin sheath

cyclic-nucleotide
phosphodiesterase activity

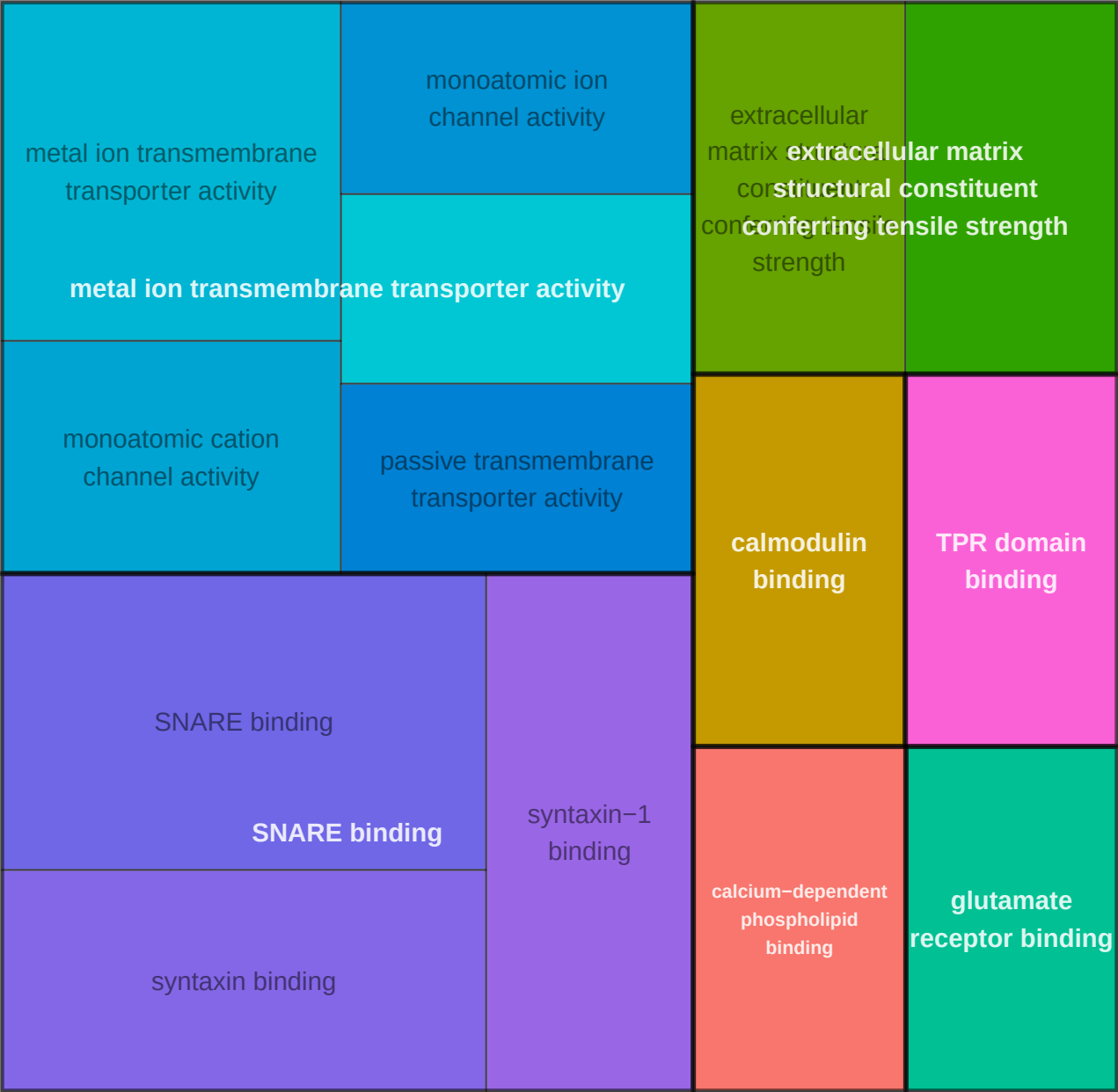
Oligo-3_up_AA CC



Oligo-3_up_AA BP



Oligo-3_up_AA MF



Oligo-3_up_AA, Oligo-3_up_EA CC

actin-based cell projection

actin-based cell projection

filopodium

Oligo-3_up_AA, Oligo-3_up_EA BP



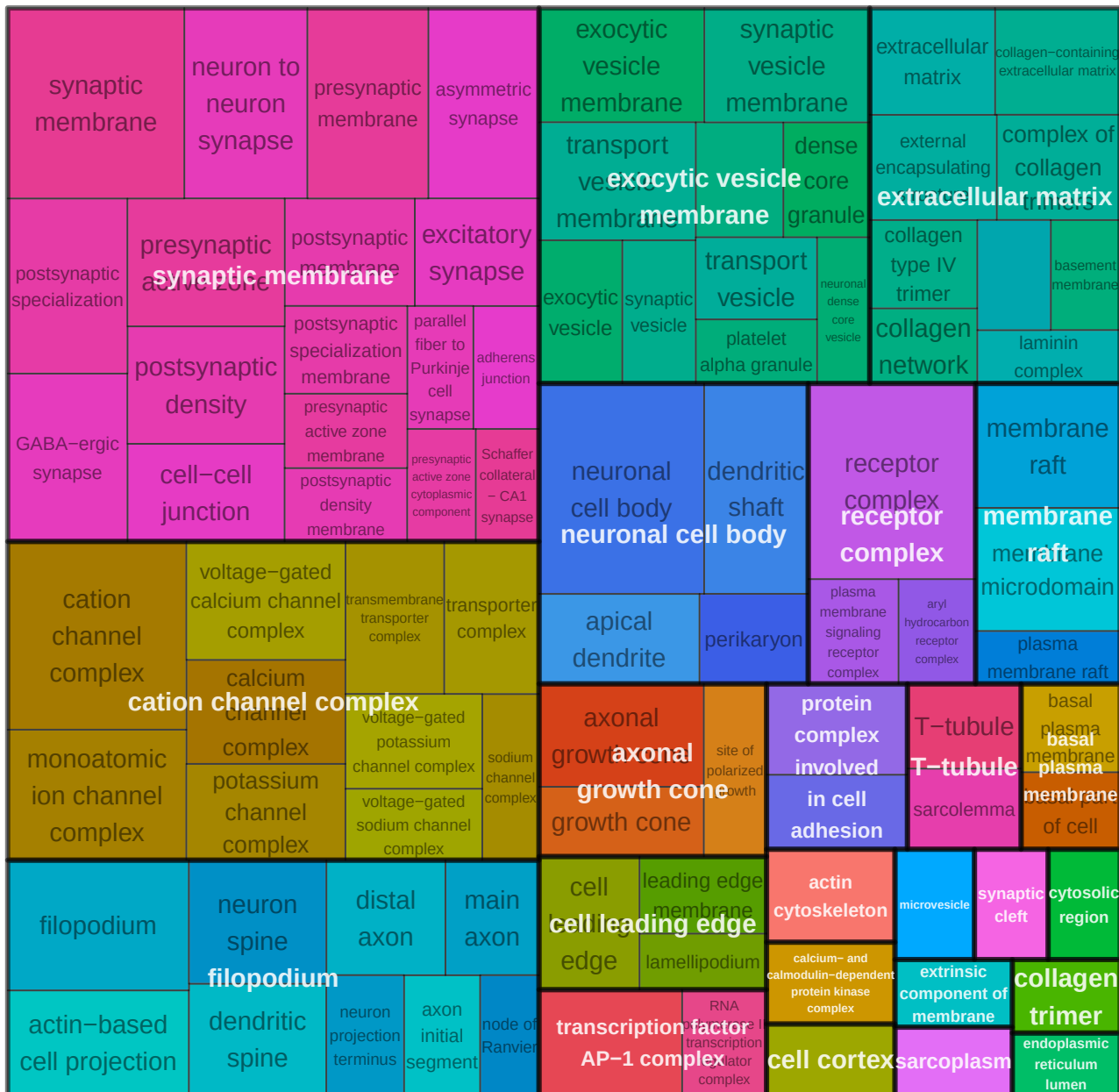
Oligo-3_up_AA, Oligo-3_up_EA MF

14-3-3 protein binding

mRNA 3'-UTR AU-rich region binding

R-SMAD binding

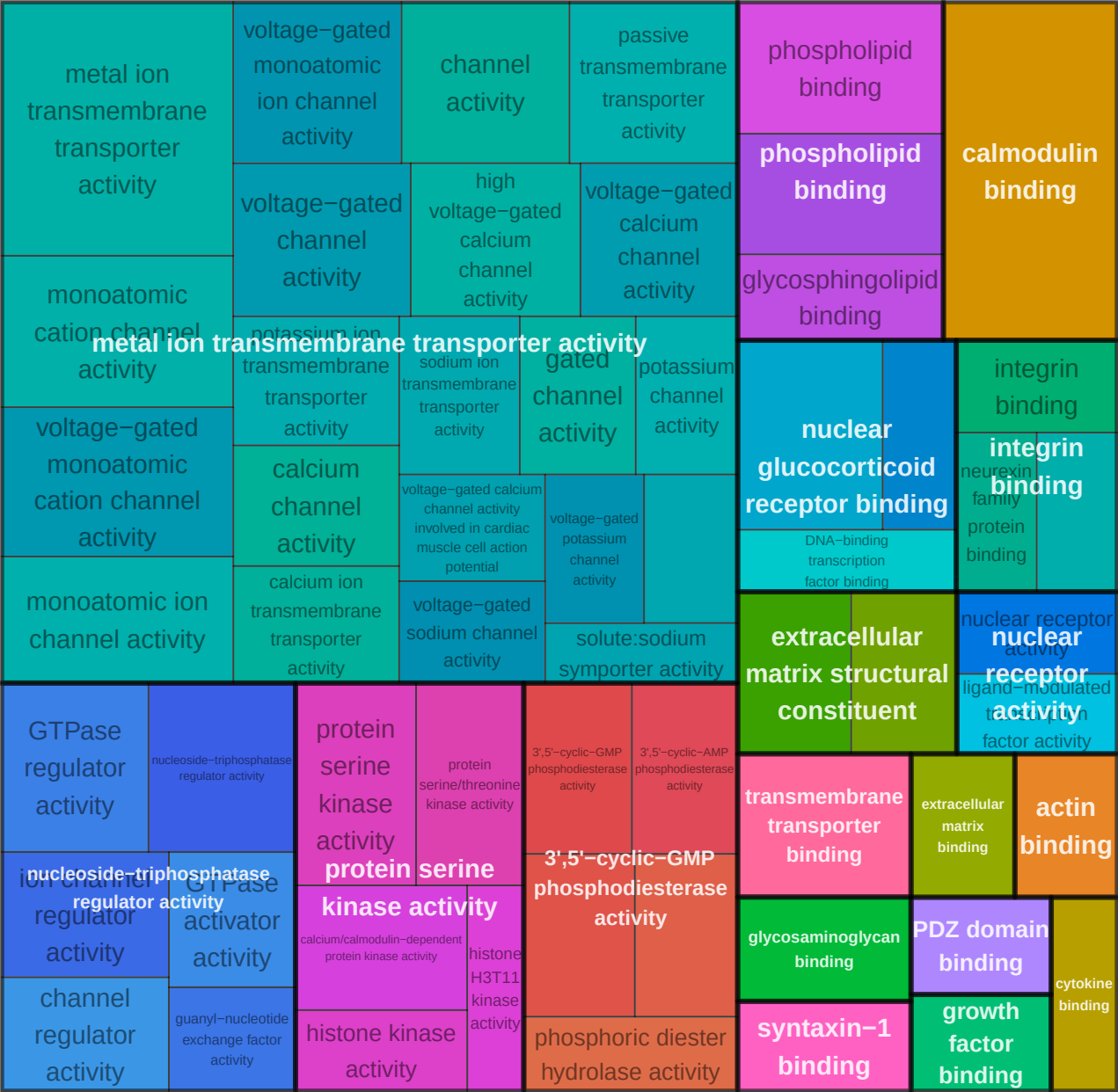
Oligo-3_up_ALL CC



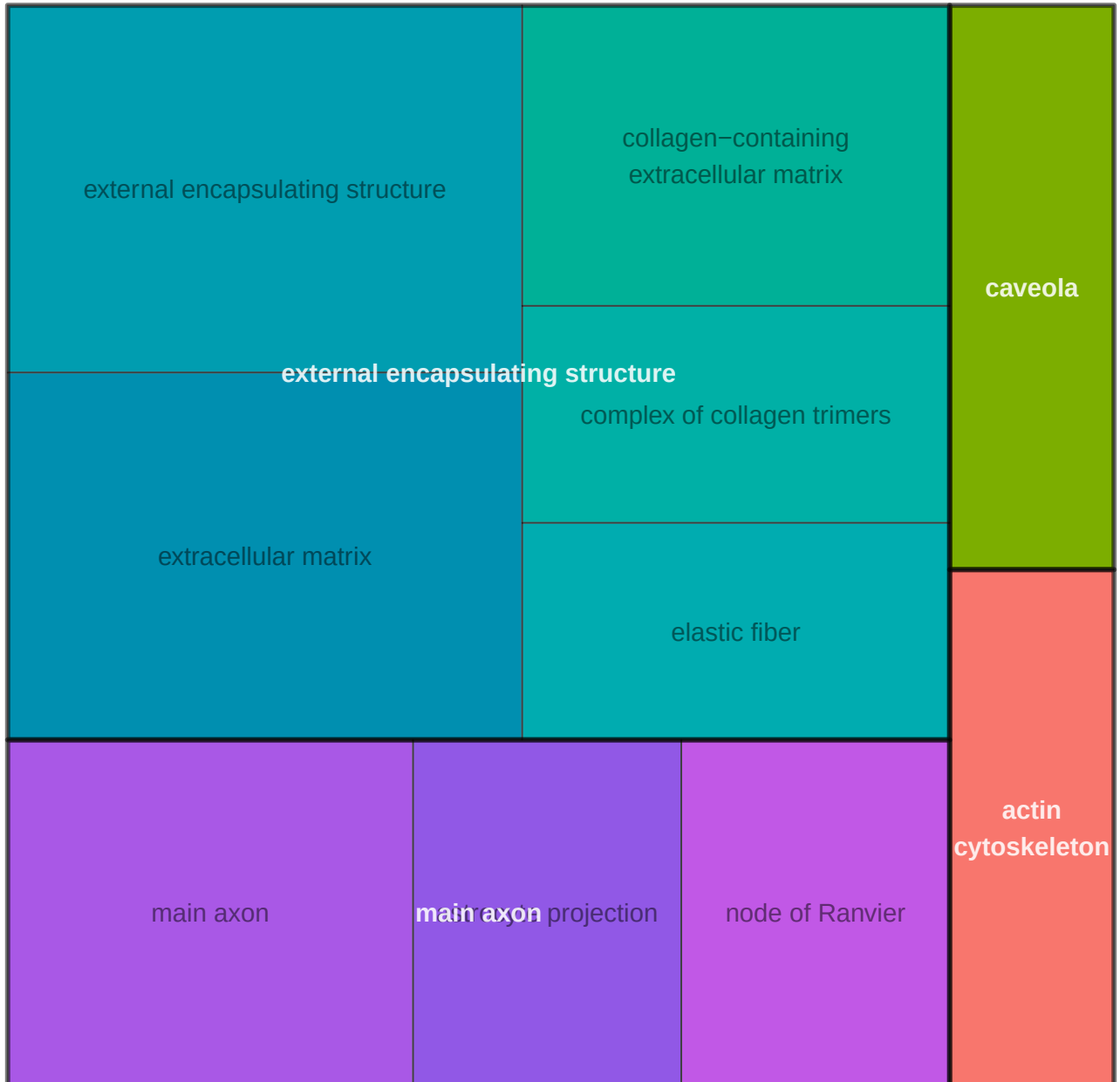
Oligo-3_up_ALL BP



Oligo-3_up_ALL MF



Oligo-3_up_EA CC



Oligo-3_up_EA BP



Oligo-3_up_EA MF

extracellular matrix structural constituent

growth factor binding

Oligo-5_down_ALL CC

collagen type IX trimer

FACIT collagen trimer

FACIT collagen trimer

recycling endosome

Oligo-5_down_ALL MF



Oligo-5_up_ALL CC

membrane microdomain

membrane microdomain

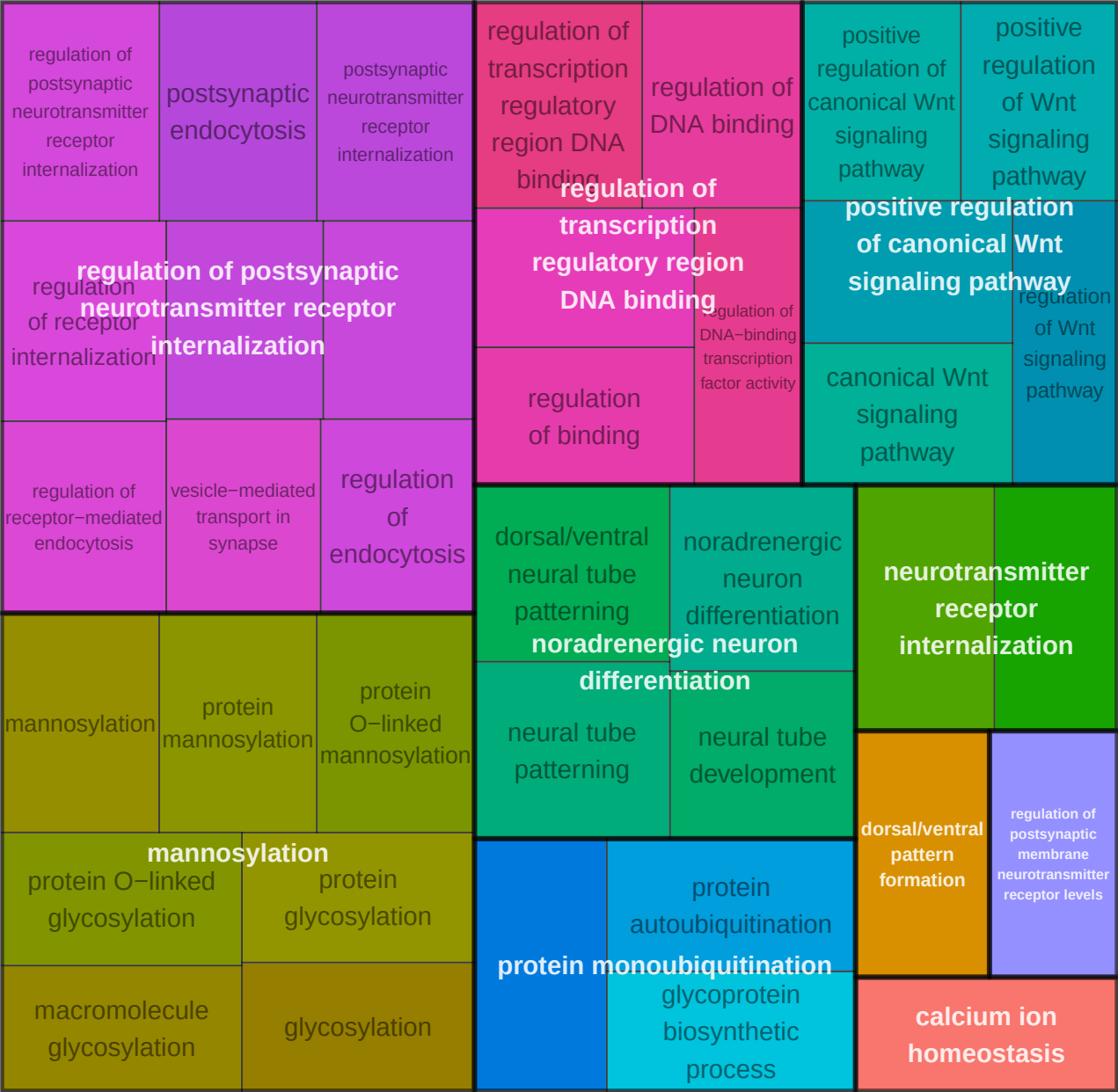
membrane raft

immunological synapse

Oligo-5 up ALL BP



OPC-4_down_AA BP



OPC-4_down_AA MF

dolichyl-phosphate-mannose-protein
mannosyltransferase activity

**dolichyl-phosphate-mannose-protein
mannosyltransferase activity**

mannosyltransferase activity

beta-catenin binding

Vasc-Endo_up_EA CC



Vasc-VLMC_down_AA CC

filopodium tip

filopodium tip

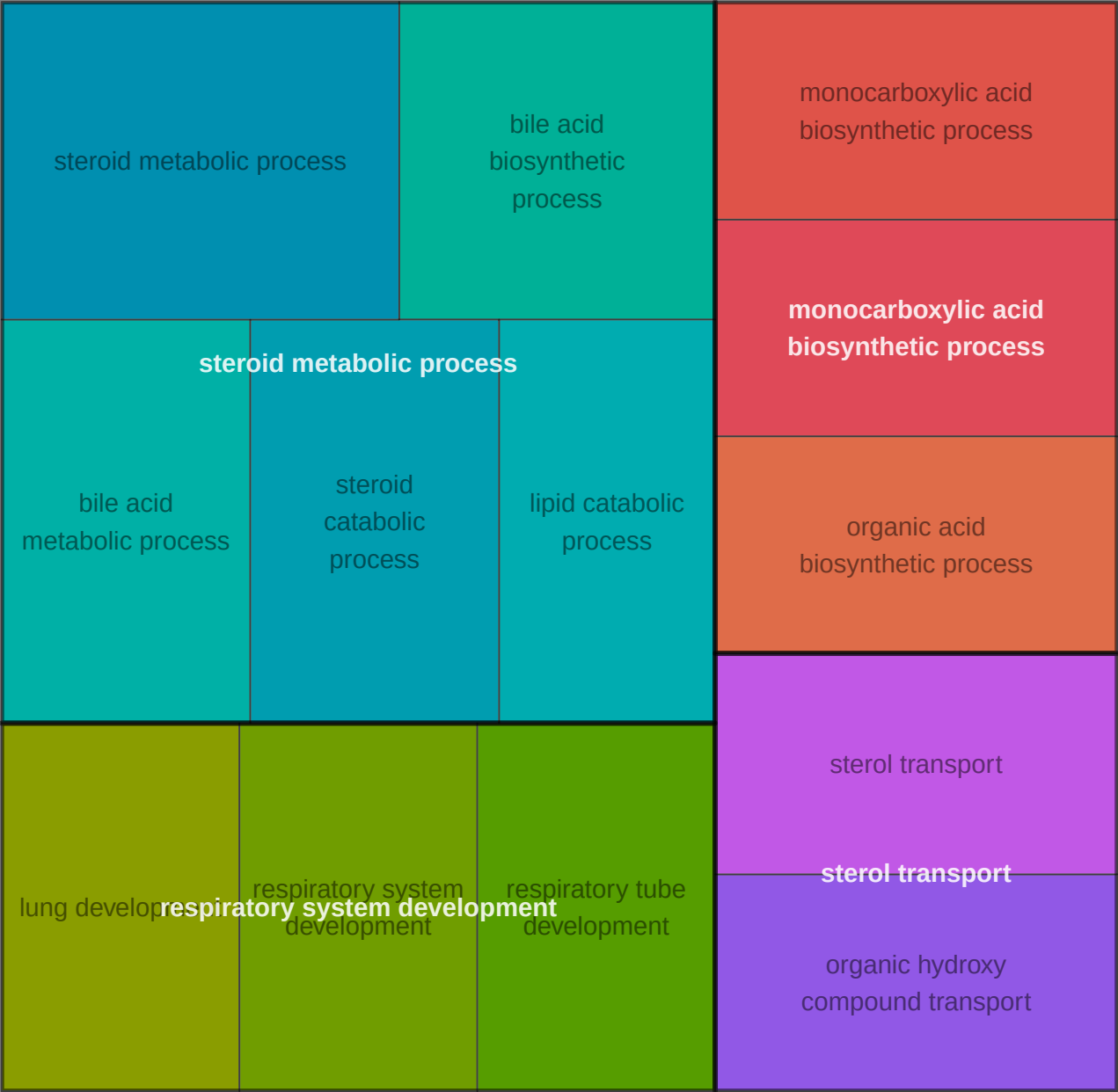
filopodium

actin-based
cell projection

**nuclear
membrane**

perinuclear endoplasmic reticulum

Vasc-VLMC_down_AA BP



Vasc-VLMC_down_ALL MF

